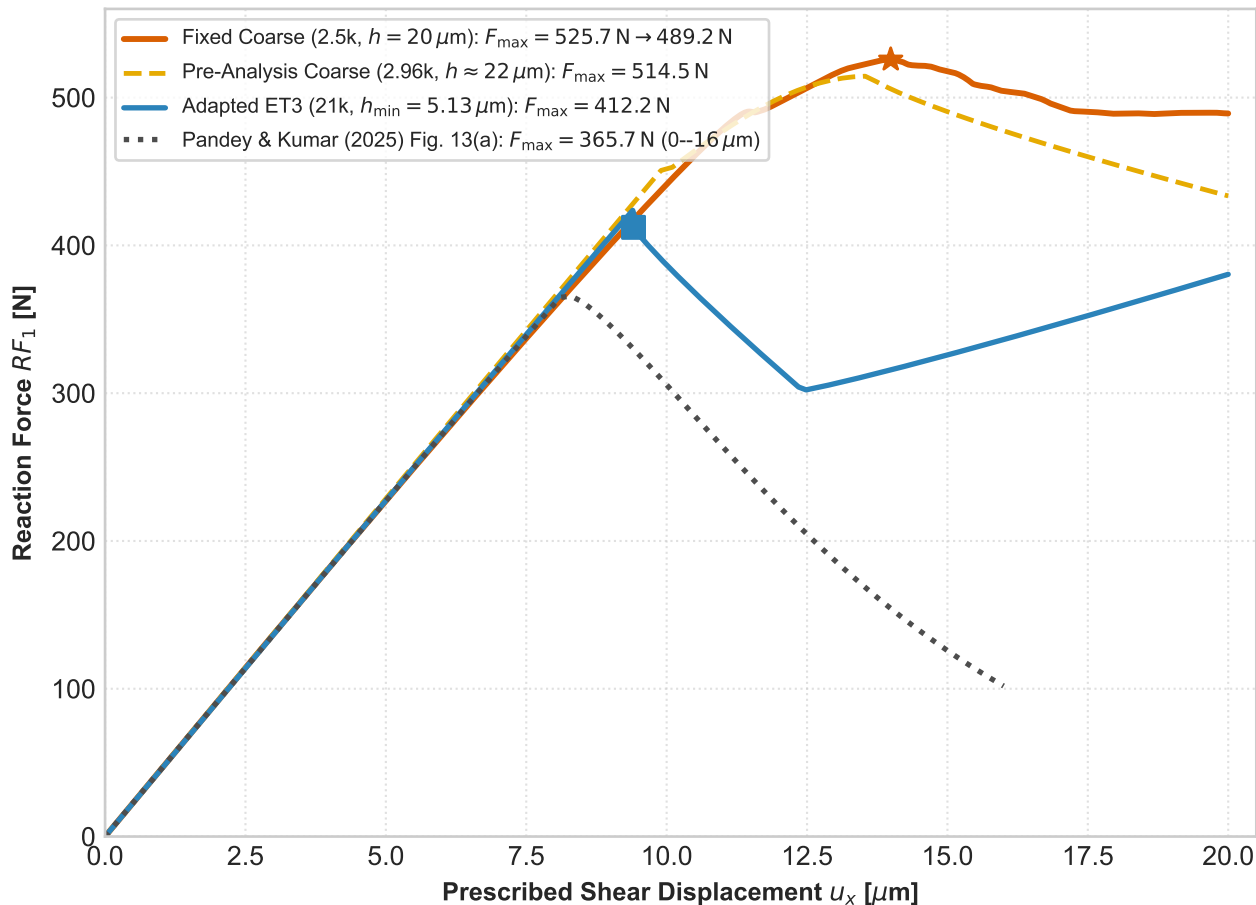
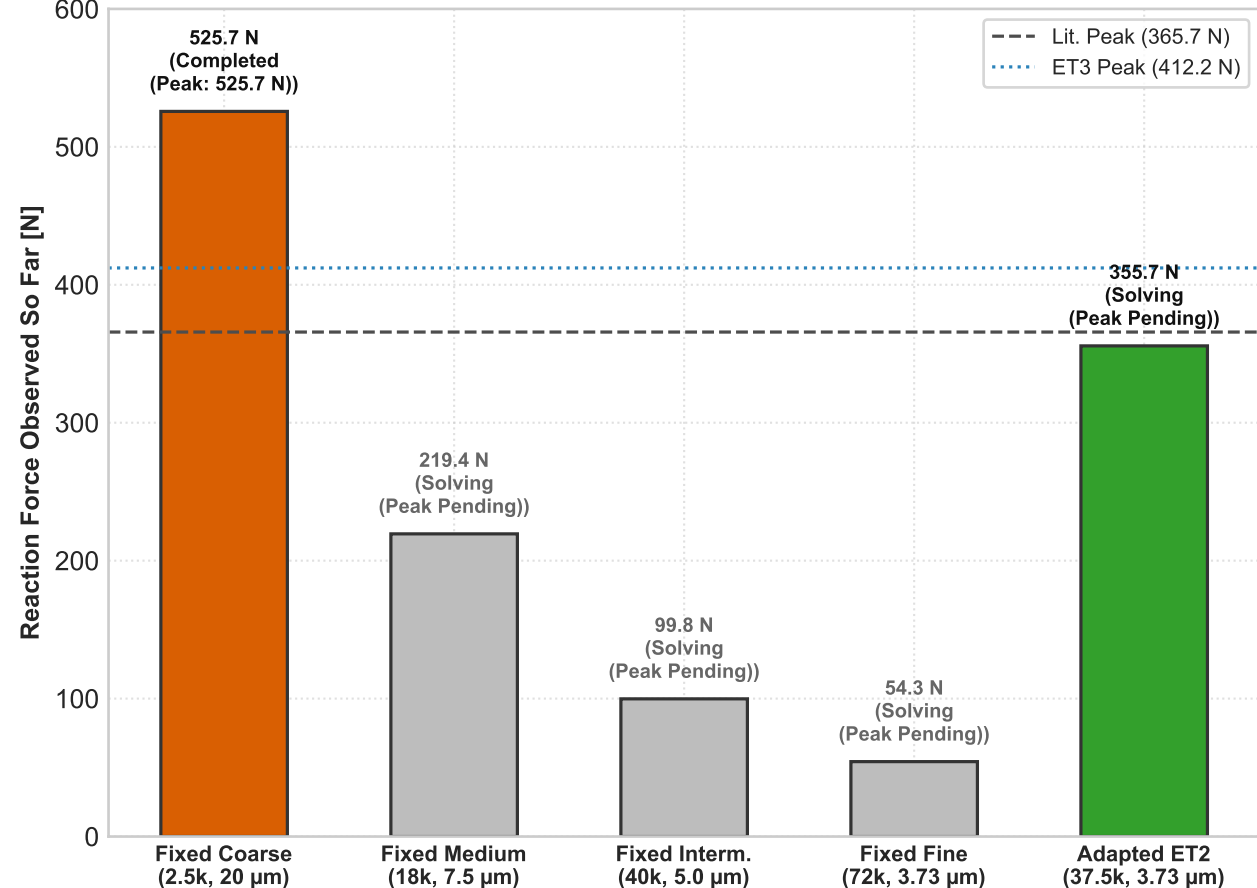
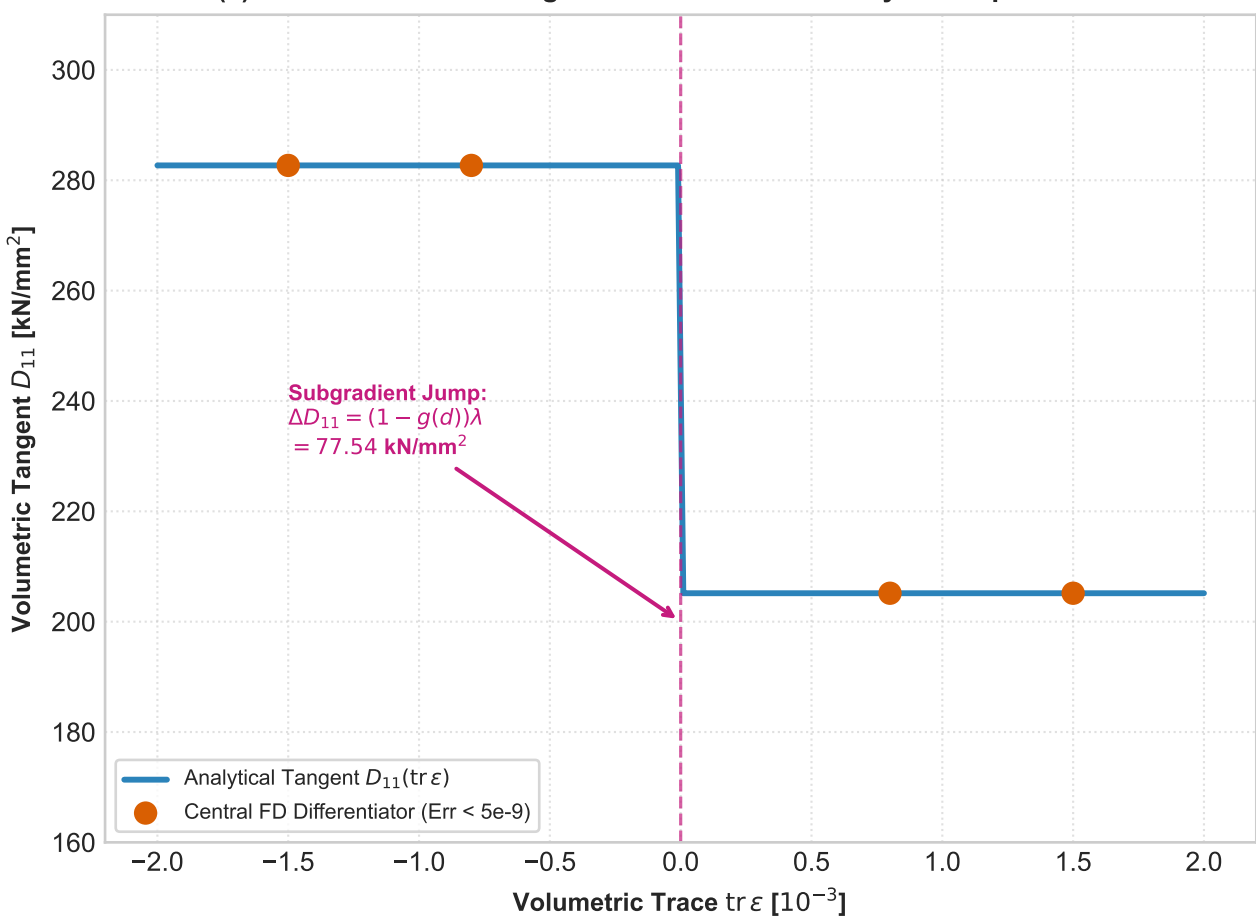


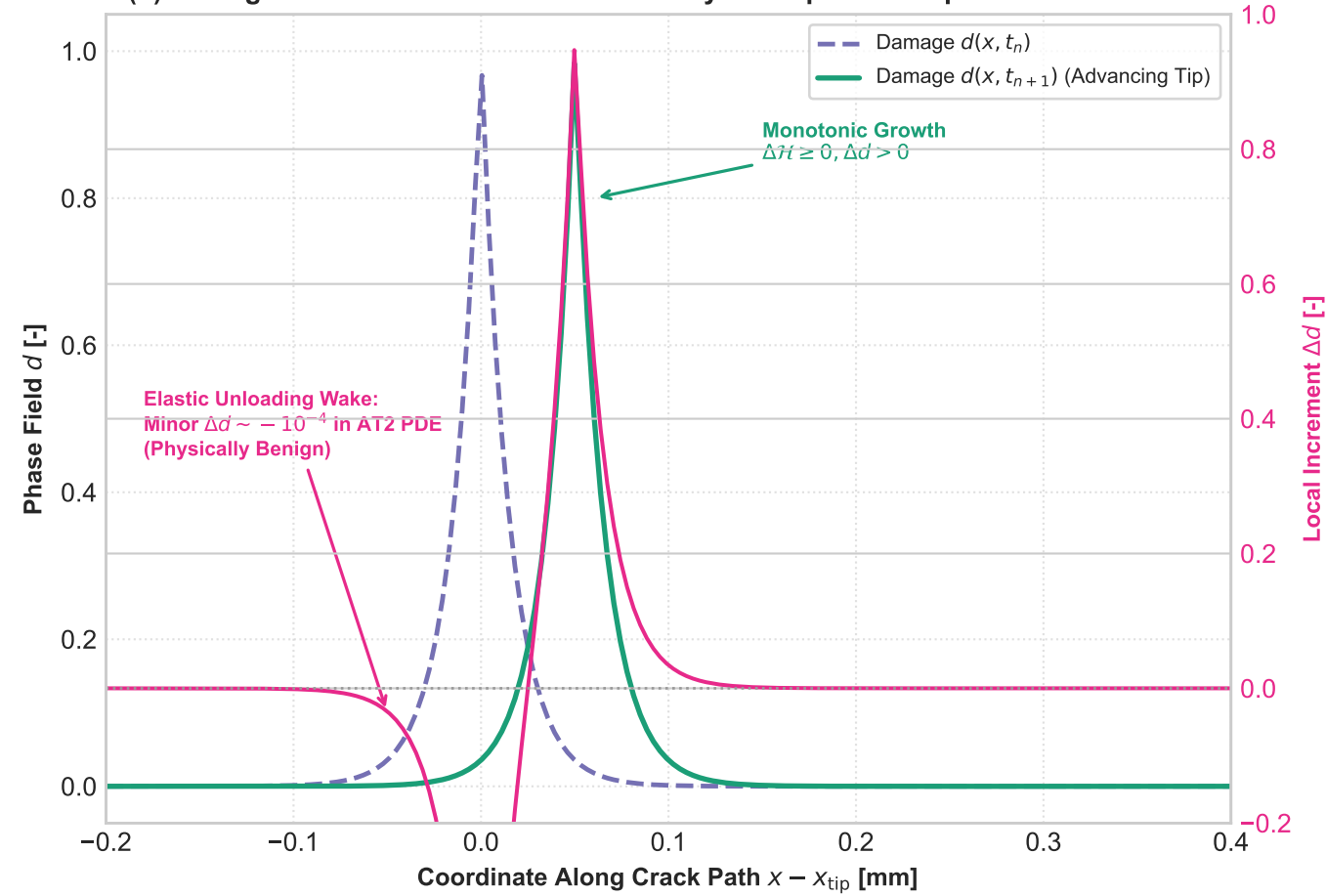
(a) Mode-II Fracture Response: Fixed Coarse vs Adapted vs Literature



(b) Gate M2-1B Evidence Status: Peak Force Evaluation in Progress

(c) UEL Constitutive Tangent: Piecewise Continuity & Jump at $\text{tr } \varepsilon = 0$ 

(d) Damage Field Kinetics: Monotonic History vs Elliptic PDE Spatial Redistribution



(e) Layer 2 Indicator Feasibility & Sizing Bounds (Proposed Architecture)

MULTI-FIELD INDICATOR: $\eta_K = \max(\eta_{\sigma, K}, \eta_{d, K})$

- Stress Discretization Error: Drives early elastic singularity refinement
- Damage Gradient Indicator: Preserves crack wake resolution
- Prevents Coarsening Defect: When $d \rightarrow 1$, $\sigma \rightarrow 0$; η_d keeps channel fine!
- Multi-Criterion Normalization: Dimensionless max avoids arbitrary fixed weights

PHYSICAL SIZING SAFEGUARDS & MESH GRADATION

- Lower Sizing Bound: $h_{\min} = l_0/4 = 3.75 \mu\text{m}$ (prevents over-refinement)
- Upper Sizing Bound: $h_{\max} = 25 \mu\text{m}$ (far-field computational efficiency)
- Sizing Gradient Limiter: $|\nabla h| \leq 0.30$ (smooth transition, aspect ratio < 1.8)
- Status: Proposed controller design parameters (subject to reference validation)

(f) Systematic Fixed-Reference to Adaptive Decision Gates

GATE M2-1B: FIXED-MESH FRACTURE REFERENCE CONVERGENCE

Solve 4-tier uniform suite (2.5k $\rightarrow$ 18k $\rightarrow$ 40k $\rightarrow$ 72k) to complete fracture.
Establish true physical fixed-mesh peak force & trajectory without remeshing.

GATE M2-4: ADAPTIVE-REMESHING FIDELITY & EFFICIENCY

Compare adapted solutions (ET3, ET2) against verified fixed-mesh reference.
Quantify gap closure, error norms, DOF savings, and runtime speedup.

GATE 8: GENERALIZATION TO COMPLEX FRACTURE BENCHMARKS

Deploy validated 3-layer architecture to arbitrary crack paths
(mixed-mode, branched, curving cracks) without pre-prescribing trajectories.